



First Normal Form (1NF)

Insured Customer Family Data - From most-recently-entered (top) to least-recently-entered (bottom)

----- Dependent Children -----								
Customer Code	Customer Surname	Child	Age	Child	Age	Child	Age	etc.
DENT04	Denton	Billy	3	Olivia	10			
PARK31	Park	Justin	7					
PARK31	Park	Phoebe	5					
KELLI7	Keller	Alice	4	Ivy	Newborn	Billy	9	Jonah 7

The way this data set is organized **violates First Normal Form** – not just in one way but in four separate ways! **1)** It is using the order of the records to convey meaningful information (“From most-recently-entered (top) to least-recently-entered (bottom)”). **2)** A given column doesn’t always have just one data type: “Age” seems to be numeric in some places (Olivia’s Age is 10) but string in other places (Ivy’s Age is “Newborn”). **3)** No primary key: the uniqueness of Customer Code is clearly not being enforced (if it was, PARK31 couldn’t have two records). **4)** Repeating groups exist: “Child” and “Age” repeat an indefinite number of times on each record.

An alternative design that complies with First Normal Form is:

Customer

Customer_Code	Customer_Surname
DENT04	Denton
PARK31	Park
KELLI7	Keller

Customer_Dependent_Child

Customer_Code	Child_Sequence_Number	Child_Name	Child_Age	Date_Recorded
PARK31	1	Phoebe	5	2026-01-27
PARK31	2	Justin	7	2026-01-28
DENT04	1	Billy	3	2026-02-01
DENT04	2	Olivia	10	2026-02-01
KELLI7	1	Ivy	0	2026-01-19
KELLI7	2	Alice	4	2026-01-19
KELLI7	3	Jonah	7	2026-01-19
KELLI7	4	Billy	9	2026-01-19

- Both tables are in 1NF (and indeed 5NF – fully normalized)
- Primary key of Customer table = Customer_Code
- Primary key of Customer_Dependent_Child table = { Customer_Code, Child_Sequence_Number }
- Customer_Code on Customer_Dependent_Child table is a foreign key to Customer_Code on Customer table
- Date_Recorded tells us about recent / less recent records
- In a real-life database, Child_Age would probably not be captured; Child_Birth_Date would be captured instead

Second Normal Form (2NF)

Employee_Working_Days

Employee_Code	Working_Day	Employee_Surname
WREN02	TUESDAY	Wren
WREN02	WEDNESDAY	Wren
WREN02	THURSDAY	Wren
MILL01	MONDAY	Miller
MILL01	TUESDAY	Miller

This table, which captures the days that each part-time employee works, **violates Second Normal Form**. The primary key is { Employee_Code, Working_Day }. Informally, Second Normal Form requires that each non-key attribute should be dependent on the **entire** primary key. But Employee_Surname is dependent on Employee_Code, which is a **part** of the primary key. As a result, the table fails to achieve Second Normal Form.

An alternative design that complies with Second Normal Form is:

Employee

Employee_Code	Employee_Surname
WREN02	Wren
MILL01	Miller

Employee_Working_Days

Employee_Code	Working_Day
WREN02	TUESDAY
WREN02	WEDNESDAY
WREN02	THURSDAY
MILL01	MONDAY
MILL01	TUESDAY

- Both tables are in 2NF (and indeed 5NF – fully normalized)
- Primary key of Employee table = Employee_Code
- Primary key of Employee_Working_Days table = { Employee_Code, Working_Day }
- Employee_Code on Employee_Working_Days table is a foreign key to Employee_Code on Employee table



Third Normal Form (3NF)

Product

Product_ID	Product_Name	Product_Subcategory	Product_Category
103	400-Piece Autumn Foliage Jigsaw	JIGSAWS	PUZZLES
104	250-Piece Magic Bookshop Jigsaw	JIGSAWS	PUZZLES
105	Standard Rubik's Cube	MECHANICAL-PUZZLES	PUZZLES
106	Kendrick Koala	CUDDL-TOYS	TOYS
107	Adventurer Walkie Talkie Set	ELECTRONIC-TOYS	TOYS

This table, which records information about products, **violates Third Normal Form**. The primary key is Product_ID. Informally, Third Normal Form requires each non-key attribute to be dependent on the key, the whole key, and nothing but the key. It's true that Product_Category is dependent on the whole key (Product_ID), but it's equally true that Product_Category is dependent on something that isn't the key: Product_Subcategory. As a result, the table fails to achieve Third Normal Form.

An alternative design that complies with Third Normal Form is:

Product

Product_ID	Product_Name	Product_Subcategory
103	400-Piece Autumn Foliage Jigsaw	JIGSAWS
104	250-Piece Magic Bookshop Jigsaw	JIGSAWS
105	Standard Rubik's Cube	MECHANICAL-PUZZLES
106	Kendrick Koala	CUDDL-TOYS
107	Adventurer Walkie Talkie Set	ELECTRONIC-TOYS

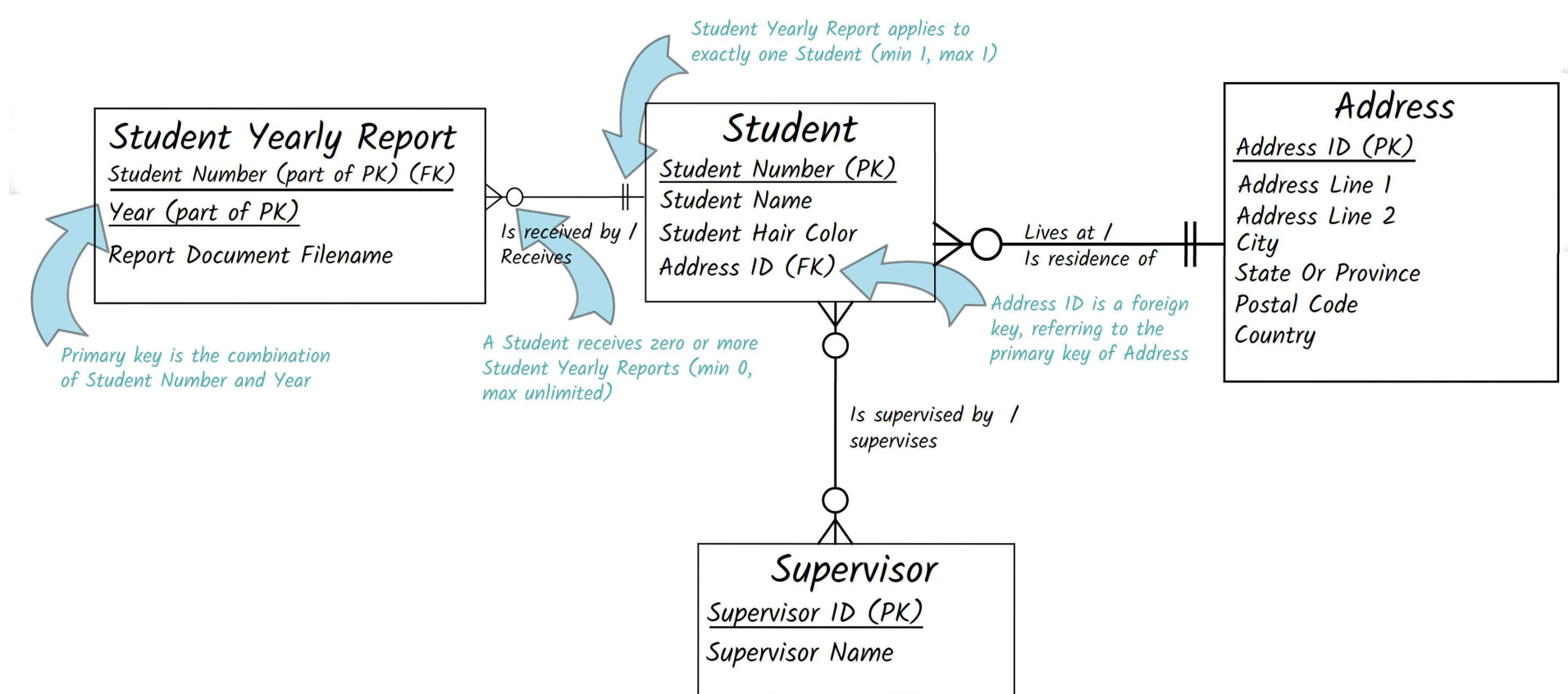
Product_Subcategory

Product_Subcategory	Product_Category
JIGSAWS	PUZZLES
MECHANICAL-PUZZLES	PUZZLES
CUDDL-TOYS	TOYS
ELECTRONIC-TOYS	TOYS

- Both tables are in 3NF (and indeed 5NF – fully normalized)
- Primary key of Product table = Product_ID
- Primary key of Product_Subcategory table = Product_Subcategory
- Product_Subcategory column on Product table is a foreign key to Product_Subcategory column on Product_Subcategory table

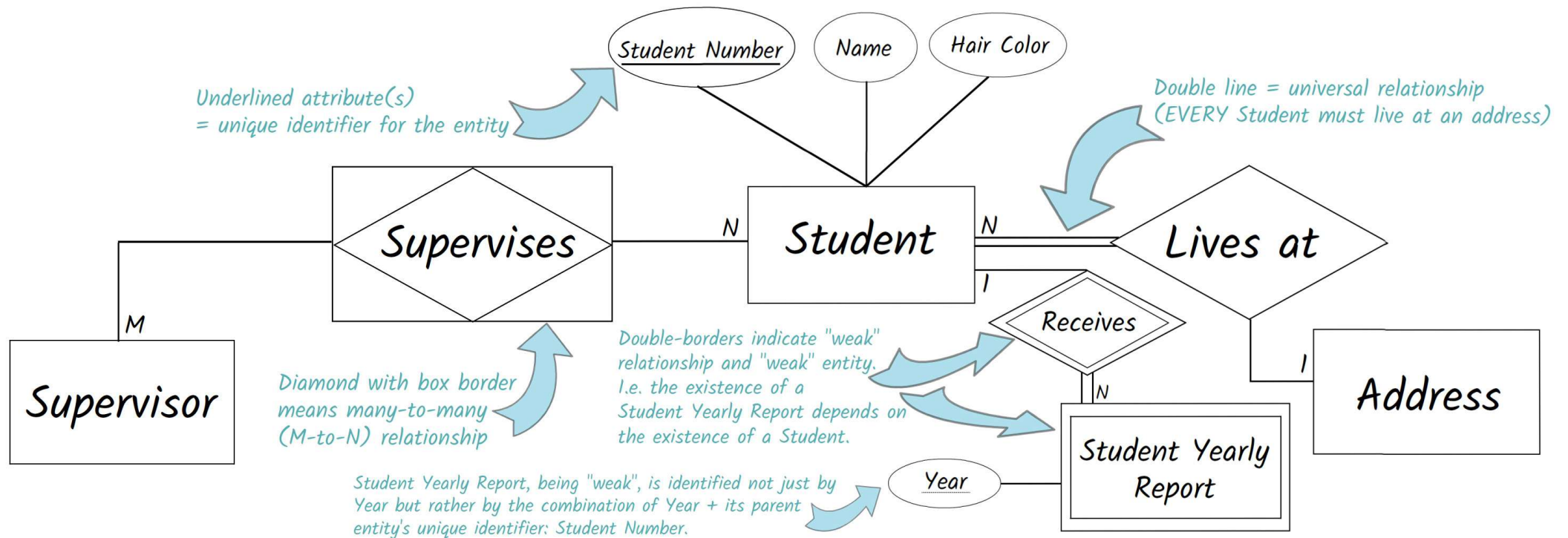


Entity Relationship Diagram (ERD): Crow's Foot Notation



- A given *Student* receives zero or more *Student Yearly Reports* (i.e. one for each year they complete). A given *Student Yearly Report* is received by exactly one *Student*.
- A given *Student* lives at exactly one *Address*. A given *Address* is the residence of zero more more *Students*.
- A given *Student* has zero or more *Supervisors*. A given *Supervisor* supervises zero or more *Students*.

Entity Relationship Diagram (ERD): Chen Notation



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- A given *Student* lives at exactly one *Address*. A given *Address* is the residence of zero more more *Students*.
- A given *Student* has zero or more *Supervisors*. A given *Supervisor* supervises zero or more *Students*.